Problem 4.

i)

$$\text{LHS} = \int_0^\infty e^{-st}\, e^{-bt} f(t)\, dt = \int_0^\infty e^{-(s+b)t} f(t)\, dt$$

$$\text{RHS} = F(s+b) = \int_0^\infty e^{-(s+b)t} f(t)\, dt$$

$$\therefore \text{LHS} = \text{RHS}$$

ii)

$$\text{LHS} = \int_0^\infty e^{-st} f(at)\, dt$$

Assume $u = at$, $\dfrac{du}{dt} = a$

$$\text{LHS} = \int_0^\infty e^{-(s/a)\, u} f(u) \times a^{-1}\, du$$

$$\text{RHS} = a^{-1} F(\frac{s}{a}) = a^{-1} \int_0^\infty e^{-(s/a)\, t} f(t)\, dt$$

$$\therefore \text{LHS} = \text{RHS}$$

This solution was submitted by **Jon**.

Published and refined by **Jon**.

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